

system. They have enabled the development and upkeep of numerous websites and web applications. This pervasiveness exemplifies the impact that open source has had on global connectivity.

- **Adoption by corporations:** Large corporations have recognised the value of open source software and implemented it in their operations. Many have established open source offices and become active contributors to open source projects.
- **Impact on emerging technologies:** Open source software has shaped emerging technologies such as artificial intelligence, blockchain, and edge computing. The availability of open source frameworks and libraries has accelerated the development of these technologies and made them more accessible to a wider range of developers.
- **Educational institution accessibility:** Open source software has been adopted by educational institutions ranging from primary schools to universities. Its low cost, adaptability, and educational licences make it an excellent choice for students and educators. This ensures that the next generation of developers understand the principles and practices of open source.
- **Improved user experience:** Open source projects have made investments to improve the user experience. Many open source applications now have user friendly interfaces, intuitive navigation, and user-centric design. This emphasis on user experience has increased the appeal of open source software to a broader audience.
- **Better licensing models:** The evolution of various licensing models has provided open source projects with greater flexibility and clarity. Licences such as the GNU General Public License (GPL), the MIT License, and the Apache License provide different options for sharing and modifying code. These licences have contributed to a better understanding of the legal landscape surrounding open source software.
- **Strong ecosystem of support:** A robust support ecosystem is often beneficial to open source software. Companies, communities, and individuals provide support and consulting services to ensure that users have access to the help they require. This infrastructure improves the dependability of open source solutions.

Open source software is a dynamic and enduring force in the world of technology due to its diverse ecosystem, commitment to security, global collaboration, and adaptability to various industries. However, the open source journey has not been without its set of challenges.

Drawbacks of OSS

- **Lack of structured support:** One of the primary disadvantages of OSS is the lack of structured, guaranteed support. Users frequently rely on community forums, user

documentation, or volunteer help. While this approach has the potential to be extremely effective, it may not be appropriate for organisations or individuals with mission-critical applications. Downtime and operational challenges can result from a lack of timely and consistent support.

- **Problems with compatibility:** Integrating various open source components can lead to compatibility issues. Different versions of software libraries may be used in different projects, and certain configurations may be required. It can be difficult to ensure that these components work well together, especially when combining multiple open source tools.
- **Licensing complexity:** Open source licences can be complex and varied. This variety can cause confusion and legal issues for users who do not fully understand the implications of different licences. The possibility of inadvertently violating licence terms or obligations is a significant disadvantage.
- **Limited documentation:** While open source projects have made strides in providing comprehensive documentation, some still lack clear and user-friendly instructions. Inadequate documentation can prevent users from fully understanding and utilising the features of the software.
- **Learning curve:** Open source software can have a steep learning curve for newcomers. This disadvantage may deter adoption and limit the user base. This challenge may be exacerbated by a lack of user-friendly interfaces or intuitive designs.

What modern trends should focus on

Modern trends in OSS may inadvertently overlook several critical aspects that can impact the ecosystem's growth and success.

- **Commercial viability:** Some open source projects struggle to find a viable business model, potentially limiting their long-term viability and discouraging contributors from investing their time and effort. Open source initiatives need to focus on fostering commercial success.
- **Accessibility:** Open source software is widely available, but it might not reach underserved communities or those with limited internet access. The aim should be to bridge the digital divide and make open source solutions more widely accessible.
- **Design and usability:** Many open source projects prioritise functionality over design and usability. Modern design trends should emphasise user-centric design, improving user experience (UX) and user interface (UI), which can broaden open source software's appeal.
- **Marketplace integration:** Contemporary trends should explore ways to incorporate open source software into larger marketplaces and ecosystems. This integration can make it easier for a broader user base to discover and adopt open source solutions.